
Drift–Utility Dissociation in Time-Series Foundation Model Fine-Tuning: Methodology and a Moirai Case Study

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We contribute a **reusable diagnostic methodology** for assessing representa-
2 tion drift during fine-tuning of time-series foundation models: a value-gate
3 screening criterion, controlled-comparison design with frozen-encoder/random-
4 init/layer-wise-unfreeze controls, task-orthogonal probe protocol, LoRA-Large
5 LR recipe, uni2ts library patch, and CUDA-deterministic reproducibility proto-
6 col. We demonstrate the methodology on a **Moirai-ETT case study** (the one
7 backbone×dataset family where the value-gate passes). The load-bearing empiri-
8 cal signals are: (1) the **task-orthogonal probe asymmetry** ($\Delta R^2 < 0$ on positive-
9 floor probes, $\Delta R^2 > 0$ on the trained head; 10/10 seeds on ETTh2 and ETTm2,
10 where ETTm2 is a gate-failing cross-domain comparison confirming the mech-
11 anism generalises), operationalising “task-specific shift”; and (2) **layer-wise lo-**
12 **calisation** (top-3 layers suffice for $\Delta R^2 > 0$ but lower layers required for task re-
13 covery; 10/10 seeds). Fine-tuning imposes geometric cost (CKA 0.949→0.52 as
14 $n:500 \rightarrow 10k$) with protocol-dependent task outcome—harmful at $n \leq 1k$ (+14%
15 forgetting), improving at $n=10k$ under early-stopped checkpointing. Five non-
16 Moirai backbones gate-fail on all nine ETT cells tested; Moirai itself gate-fails on
17 Traffic, Exchange, and Electricity, narrowing the gate-pass to ETTh2 and confirm-
18 ing that corpus composition drives coverage (§5.1). **Practitioner takeaway:** Use
19 LoRA with $10\times$ LR reduction on larger Moirai variants before escalating rank;
20 apply the value-gate before investing in drift diagnosis.

21 1 Introduction

22 Moirai Woo et al. [2024] is a foundation time-series model trained on billions of time points that
23 achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune
24 it on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with lim-
25 ited labelled data. We present a Moirai case study on ETT forecasting benchmarks that characterises
26 what happens to pre-trained representations during fine-tuning.

27 **Representation drift during fine-tuning.** We contribute a systematic *diagnostic methodology* for
28 characterising representation drift—progressive overwriting of pre-trained features—during fine-
29 tuning of time-series foundation models, and demonstrate it on Moirai (Small 13.8M, Base 91M,
30 Large 311M) as the single backbone family where pre-trained features pass our value-gate on ETT
31 benchmarks. Matched-protocol attempts to replicate on three non-Moirai backbones (Chronos-T5-
32 Small Ansari et al. [2024], TimesFM-2.5-200M Das et al. [2024], MOMENT-1-base Goswami et al.
33 [2024]) gate-fail on all nine ETT×backbone cells tested (7 original + 2 new Chronos capacity cells);
34 we report this as a substantive finding about backbone-specific ETT pre-training coverage (§4, Ap-
35 pendix A). Catastrophic forgetting during fine-tuning is well-studied in continual learning Kirk-

patrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]; our contribution is reusable diagnostic methodology, demonstrated on a well-controlled Moirai case study. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the “previous task” is the pre-training objective.

Distinguishing drift from overfitting. Representation drift degrades zero-shot performance while training loss decreases (not overfitting); the frozen-encoder control (CKA=1.000, same data/epochs) isolates encoder weight updates as the cause.

On ETTh2 (gate-pass: Moirai outperforms Linear by 28–45%, $R^2_{\text{task}}(\text{PT})=+0.279$), fine-tuning degrades MSE by 10–16% at $n=500$ and resolves by $n=2,000$. ETTh1 and ETTm2 gate-fail at the standard evaluation protocol and are included as cross-domain comparisons (Table 2). On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. Time-series drift directly harms distributional predictions for anomaly scoring and probabilistic forecasting. Consequences depend on pre-training value: harmful on ETT in low-data regime ($n \leq 1\text{k}$), benign at large n once replacements are learned. *Counterfactual without the value-gate:* Chronos-T5-Small on ETTh2 gives ZS MSE=0.304 vs. Linear 0.213 (gate-fail: ZS underperforms by 43%). A practitioner who skipped screening would fine-tune Chronos, observe CKA dropping, diagnose “drift,” and apply LoRA/EWC to preserve features that had *no advantage over Linear*—wasted effort. The value-gate prevents this: drift diagnosis is meaningful only when pre-trained features demonstrably outperform a trivial baseline.

Scope. The empirical case study is narrow by design: one gate-passing backbone (Moirai) on one clean setting (ETTh2, $n=500\text{--}10\text{k}$). We do not claim a universal phenomenon; we claim reusable methodology and one well-controlled case study illustrating its application. The methodology (value-gate, controls, probes, protocol) transfers to any backbone; the specific numeric findings require replication.

Experimental approach. We deploy the diagnostic methodology across three Moirai sizes (Small/Base/Large) on ETT forecasting Zhou et al. [2021] (ETTh2 gate-pass; ETTh1/ETTM2 cross-domain) and IoT anomaly detection (negative control; §5). Forgetting sign at $n=10\text{k}$ is protocol-dependent (−5.3% ES vs. +7.5% final-epoch); we lead on the more robust $\Delta R^2 > 0$ majority.

Contributions. Methodological contributions are offered as reusable infrastructure; empirical findings are Moirai-ETT specific and require replication.

1. **Diagnostic methodology (backbone-agnostic).** Value-gate screening criterion (ZS >20% vs. Linear); controlled-comparison design (frozen-encoder, random-init, layer-wise unfreeze controls); LoRA-Large $10\times$ LR rescue recipe (cross-dataset validated); uni2ts PackedStdScaler patch; and CUDA-deterministic per-seed protocol (Appendix H).
2. **Empirical validation (Moirai-ETT case study).** Task-orthogonal probe asymmetry ($\Delta R^2 < 0$ on positive-floor probes, $\Delta R^2 > 0$ on trained head; 10/10 seeds on ETTh2/ETTM2); layer-wise localisation (top-3 layers suffice for $\Delta R^2 > 0$, lower layers for task recovery; 10/10 seeds); corpus-coverage confirmation (Chronos gate-passes M4-Monthly at 84.5% while gate-failing ETT at −43%; Moirai gate-fails on Traffic/−38%, Exchange/−21%, Electricity/−28%; §5.1).

2 Background and Problem Setup

Catastrophic forgetting in fine-tuning. Catastrophic forgetting during fine-tuning is extensively studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]. Kumar et al. [2022] show full fine-tuning can distort pre-trained features relative to linear probing; Andreassen et al. [2023] characterize OOD robustness decay throughout fine-tuning. Our contribution is a systematic *characterization* for time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across domains with contrasting pre-training value.

Time-series foundation model adaptation. Recent work on adapting pre-trained time-series models includes probabilistic fine-tuning (Lag-Llama Rasul et al. [2024]), generative pre-training with task-specific heads (Timer Liu et al. [2024]), LLM reprogramming (Time-LLM Jin et al. [2024]),

and cross-task transfer (One Fits All Zhou et al. [2023]). These works develop *adaptation recipes*; our contribution is orthogonal: a *diagnostic methodology* for characterising what happens to pre-trained representations during any such adaptation, applicable when the practitioner needs to decide whether drift is harmful or benign.

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ and downstream dataset $\mathcal{D}$ of size n , fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting). We measure encoder changes using three diagnostics (§3): (1) CKA between f_θ and $f_{\theta'}$ (geometric similarity), (2) linear probe R^2 onto the downstream target ($\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$), and (3) weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning destroys valuable features requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline; ETTh2 at $h=96$ passes with Moirai-Small ($R_{\text{task}}^2(\text{PT}) = +0.279$). ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R_{\text{task}}^2(\text{PT}) = -0.41$ and -0.56 respectively; Table 2); they are included as cross-domain comparisons to characterise drift behaviour outside the gate-pass regime. Domains that gate-fail serve as negative controls. The 20% threshold is calibrated to ETT short-horizon benchmarks where linear baselines are surprisingly competitive Zeng et al. [2023]: a model must substantially outperform a lookback-96 linear regression before drift can be *harmful* (rather than irrelevant). The threshold is a practical screen, not a universal constant; we report $R_{\text{task}}^2(\text{PT})$ values for each cell (Table 2) so readers can apply alternative thresholds.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse). $\text{Forg.} > 0$ indicates catastrophic forgetting (degraded performance vs. pre-training baseline).

3 Fine-Tuning Protocol

Moirai for forecasting and anomaly scoring. Moirai Woo et al. [2024] predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at $h=96$ and $h=192$ (Appendix C). For IoT the anomaly score is the fraction of timesteps outside the $1-\alpha$ CI ($\alpha=0.05$) over a 32-timestep horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds supervised contrastive loss Khosla et al. [2020]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by time-series position) serve as SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n=500$ (10 seeds) and is swept across $n \in \{500, 1k, 2k, 5k, 10k\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

Deterministic training protocol. All $n=10k$ per-seed results use CUDA-deterministic training: `CUBLAS_WORKSPACE_CONFIG=:4096:8`, `torch.backends.cudnn.deterministic=True`, seeded DataLoader workers, and fixed PYTHONHASHSEED. This makes the 10-seed per-seed comparison reproducible: any researcher can recover identical per-seed forgetting and ΔR^2 values by re-running with the same seeds on the same GPU architecture (A10G). Full seed list, per-seed values, and hardware specification are in Appendix H. Cross-hardware reproducibility (e.g., A100 vs. A10G) has not been empirically verified; the deterministic guarantee applies within the same GPU architecture (A10G, as used in this paper).

uni2ts PackedStdScaler patch.¹ Any Moirai fine-tuning reproduction of this paper requires the patch.

Representation diagnostics. We track representation change with three metrics: (1) CKA Kornblith et al. [2019] between pre-trained and fine-tuned encoder representations (geometric similarity); (2) linear probing R^2 (Ridge, MLP, Linear-Forecaster heads on frozen representations, predicting forecasting target; we report $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$; absolute $R^2(\text{FT})$ remains negative, so ΔR^2

¹We patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd; a differentiable `torch.where` replaces the in-place write. All results use the patched scaler (released with our code). See Appendix B for details.

quantifies relative improvement over pre-trained linear decoding); and (3) weight drift (ℓ_2 distance from pre-trained parameters).

4 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks Zhou et al. [2021] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 4.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

4.1 Experimental Setup and Value Gate

We evaluate on ETTh2 (7 features, 17k timesteps, 12/4/4 month split). Moirai-Small (13.8M) zero-shot outperforms Linear by 28–45% (ZS MSE: 0.126 at $h=96$, 0.158 at $h=192$; un-normalised gate-pass scale: 0.269 vs. 0.373 at $h=96$, 0.299 vs. 0.544 at $h=192$), passing the value gate (20% threshold). The 20% threshold is a practical screen, not a phase boundary: at 10–20%, drift diagnosis applies but marginal features reduce both harm from drift and benefit from mitigation; we recommend full probes at $\geq 20\%$ and CKA-only at 10–20%. Conditions: **B** (NLL-only), **C** (NLL+SupCon), **D** (frozen encoder), **E** (LoRA $r=8$), **F** (L2-SP), **G** (EWC). Training: $n=500$ samples, 20 epochs, AdamW ($\text{lr}=1\text{e-}4$), 10 seeds (B/C/D) or 5 seeds (E/F/G). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained/fine-tuned encoders.

4.2 Representation Drift Degrades High-Value Features

Fine-tuning degrades ETTh2 performance where pre-trained features are valuable (Appendix Table 4). Conditions B (NLL-only) and C (NLL+SupCon) degrade by 10–16% MSE (10/10 seeds positive, $p=0.002$ sign test; $d=1.23$, 95% CI [0.37, 2.01]; bootstrap $p<10^{-4}$) while training loss *decreases*—not overfitting (which raises validation loss), but representation drift. The frozen-encoder control (D, CKA=1.000, same data/epochs) eliminates drift, ruling out the training regime. Degradation scales with horizon: worse at $h=192$ (13–16%) than $h=96$ (10–11%). Contrastive loss (C) does not prevent drift: C (+10.1%) and B (+11.1%) differ by 1.0%, within seed standard deviation—a null result for the contrastive objective.

4.3 Mitigation Spectrum

We evaluate five mitigation strategies (Appendix Table 5): LoRA (E), frozen encoder (D), EWC (G), L2-SP (F), and contrastive loss (C). Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift).

LoRA achieves the best trade-off. LoRA ($r=8$, $\alpha=16$) *improves* over zero-shot at $h=96$ while maintaining high CKA (0.979) and zero weight drift. On Moirai-Large, the same recipe fails at default $\text{LR}=10^{-4}$ ($+25.7\pm 3.2\%$, 3 seeds); rank escalation does *not* rescue. Reducing LR by $10\times$ to 10^{-5} does rescue: $\text{forg.} = -8.5\pm 0.7\%$ (CKA=0.989, 5 seeds), transferring cross-dataset to ETTh1/ETTh2 at $n=500$. **Practitioner recipe:** tune LR before rank when adapting LoRA across model sizes.

EWC and L2-SP reduce but do not eliminate drift. Both improve CKA over unconstrained fine-tuning but MSE remains 18–19% above zero-shot. Frozen encoder (D) eliminates drift (CKA=1.000) but cannot adapt—a limitation when pre-trained features are insufficient.

4.4 Sample-Size Sweep: Drift–Forgetting Dynamics

Table 1 ($n\in\{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$, ETTh2) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with n (CKA from 0.949 at 500 to 0.518 at 10k), but forgetting is non-monotonic: severe at low n (+14% at 500), resolved by 2k (−2.9%), predominantly negative at 10k (−5.3±6.2%, 7/10 negative, 10-seed CUDA deterministic). The $n=5\text{k}$ variance ($+6.6\pm 10.0\%$, 7 seeds) is bimodal: 4 seeds cluster at −0.1%, 3 at +15.5% with epoch-1 overshoot (full trajectory

Table 1: Sample-size sweep on ETTh2 ($h=96$, condition B). Drift grows monotonically (CKA $\downarrow$ as $n\uparrow$); forgetting is *non-monotonic*: severe at $n\leq 1,000$, resolved at $n=2,000$, with elevated variance at $n=5,000$ (crossover regime). At $n=10,000$: CUDA 10-seed deterministic early-stopped, 7/10 negative forgetting; **10/10** $\Delta R^2 > 0$ (per-seed values in Appendix H). ΔR^2 is a *relative* signal: absolute R^2 (FT) remains negative ($R^2_{ZS} = -6.90$ for all rows). Ridge ΔR^2 at $n=500$ ($k=2$) and $n=1,000$ ($k=1$): point values only.

Samples	MSE	Forg.%	CKA	R^2 (FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148 $\pm$.008	+14.0 $\pm$ 6.7	.949 $\pm$.010	-6.79 ($k=2$)	+12 ($k=2$)	-.26 ($k=1$)
1,000	.147 $\pm$.010	+13.4 $\pm$ 9.1	.936 $\pm$.025	-6.77 ($k=1$)	+13 ($k=1$)	+.30 ($k=1$)
2,000	.126 $\pm$.008	-2.9 $\pm$ 5.9	.814 $\pm$.042	-6.51 ($k=3$)	+.39 $\pm$.35	+.73 $\pm$.75
5,000	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	-6.41 ($k=7$)	+.50 $\pm$.26	—
10,000 ^{cuda}	.122 $\pm$.008	-5.3 $\pm$ 6.2	.518 $\pm$.188	—	+.67 $\pm$.23 [†]	—

^{cuda} CUDA 10-seed deterministic (A10G); all completed, mean best epoch 4.2 $\pm$ 1.4 (Appendix H for per-seed values and protocol comparison). [†]10/10 positive; 95% Wilson CI [0.72, 1.00]. Forgetting sign: 7/10 negative (95% Wilson CI [0.40, 0.89]; protocol-dependent—see §5). $\Delta R^2 > 0$ is the robust signal.

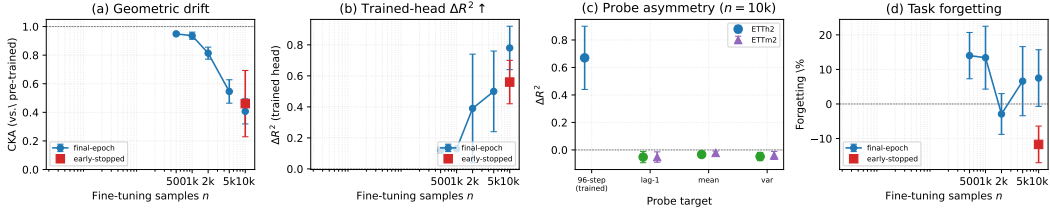


Figure 1: Cost-benefit dynamics on Moirai-Small/ETTh2 (condition B, $h=96$). (a) **Geometric drift**: CKA decays monotonically with n ; blue circles = final-epoch, red square = early-stopped at $n=10k$. (b) Trained-head Ridge ΔR^2 rises with n (same protocol markers). (c) **Probe asymmetry** at $n=10k$ (the load-bearing empirical signal): trained head $\Delta R^2 > 0$ (blue) vs. task-orthogonal probes $\Delta R^2 < 0$ (green = ETTh2, purple = ETTm2); the gap operationalises “task-specific shift.” (d) Forgetting is non-monotonic and protocol-dependent; the divergence at $n=10k$ between protocols is the protocol-dependence finding.

analysis in Appendix H.1). Linear probing shows ΔR^2 increasing with n : +0.12 ($k=2$ probe seeds) at 500 to +0.67 $\pm$ 0.23 at 10k (**10/10** seeds positive). Geometric dissimilarity (CKA $\downarrow$) and relative linear-decodability ($\Delta R^2\uparrow$) move in opposite directions.

Absolute R^2 (FT) remains negative across all n ($R^2_{ZS} = -6.90$), so ΔR^2 is a relative-improvement signal (Appendix H). Cross-dataset and model-size comparisons are detailed in §5. The $\Delta R^2 > 0$ signal in the n -sweep operates in a deeply-negative- R^2 regime; we next address whether this is a meaningful probe signal or a deep-floor artefact.

Interpretability caveat. Absolute R^2 (FT) remains negative throughout (-6.90 to -6.24); ΔR^2 is therefore a *relative* signal, not a claim about absolute decodability. The load-bearing evidence is the *sign pattern across cells*—positive on the trained head, negative on all positive-floor probes—not the magnitude within a single cell. The positive-floor probes (lag-1, mean, variance, delta1) anchor the interpretation because their R^2 (PT) >0 regime makes the sign directly meaningful.

Headline: drift is harmful in the low-data regime ($n\leq 1k$), where practitioners most need pre-trained features.

The joint trajectory is plotted in Fig. 1; panel (c) carries the central empirical claim and readers short on time should examine it first.

Task-native probe: positive-floor R^2 and $\Delta R^2 > 0$. The frozen-probe paradigm (Ridge/GBM on mean-pooled encoder outputs) gives R^2 (PT) < 0 because Moirai’s encoder outputs are optimised for its internal mixture-distribution head, not for generic linear readout. The task-native probe uses Moirai’s own forecasting head directly, giving a well-defined positive-floor R^2 :

Task-orthogonal probes (lag-1 autocorrelation, input mean, input variance) show $\Delta R^2 < 0$ on positive-floor targets ($R^2(\text{PT}) > 0$; 9–10/10 negative on ETTh2; 10/10 on ETTm2; Appendix J.1), while the trained head gives $\Delta R^2 > 0$ (10/10 on ETTh2/ETTM2). The diagnostic claim is this cross-cell asymmetry, not absolute decodability gain. ETTm2 at $n=10\text{k}$ replicates the CKA↓, $\Delta R^2 > 0$ pattern with opposite task outcome (10/10 positive Ridge ΔR^2 ; 10/10 negative forgetting); ETTh1 is a boundary case where task improves but frozen-probe ΔR^2 is 2/10 positive. Both are detailed in §5.

4.5 Layer-Wise Localisation: Probe Shift and Task Recovery Are Separable

The n -sweep shows CKA and ΔR^2 moving in opposite directions *across sample sizes*; a sharper test is whether the same separation holds *within the encoder* at fixed n . Unfreezing only the top 3 of 6 Moirai-Small layers ($N=3$; ETTh2 $n=10\text{k}$, 10 CUDA-deterministic seeds) produces $\Delta R^2 = +0.399 \pm 0.199$ (10/10 positive; 95% Wilson CI [0.72, 1.00]) with CKA=0.663 and forgetting $-6.5 \pm 4.0\%$ (9/10 negative). Full unfreeze ($N=6$, same seeds) produces $\Delta R^2 = +0.701 \pm 0.180$ (10/10 positive) with CKA=0.387 and forgetting $-6.7 \pm 6.3\%$ (6/7 negative). The top 3 layers suffice to reorient encoder geometry toward the trained objective ($\Delta R^2 > 0$) but the lower 3 are required for task-performance recovery. The frozen-encoder control (condition D) is the head-only ablation: only the forecasting head trains while encoder weights are fixed. $\Delta R^2 \approx 0$ in D (-0.048 ± 0.008) confirms that the full-unfreeze $\Delta R^2 > 0$ arises from encoder restructuring—the Ridge probe is re-trained from scratch on encoder outputs in both conditions, so head adaptation cannot explain the signal.

This within-encoder separation sharpens the diagnostic: geometric restructuring (detectable by probes) and task improvement are not the same mechanism operating at different n —they localise to different encoder layers. Per-layer CKA confirms: bottom 3 layers drift more (mean CKA=0.476) than top 3 (0.649), despite the top 3 being sufficient for $\Delta R^2 > 0$ (full analysis in Appendix M).

This layer-wise control is the cleanest demonstration of why the full diagnostic toolkit matters: without it, a practitioner observing $\Delta R^2 > 0$ at $N=3$ would conclude that fine-tuning succeeded—yet forgetting is only -6.5% (modest task benefit relative to $N=6$). Conversely, at $N=6$, forgetting is -6.7% (comparable task improvement) but the additional geometric cost (CKA 0.66→0.39) might trigger unnecessary mitigation. The diagnostic distinguishes: probes detect *reorientation* at $N=3$; task metrics require the lower layers ($N=6$) for *recovery*. Neither alone tells the full story.

5 Cross-Domain Analysis and Discussion

Per-seed cost–benefit regularity. At $n=10\text{k}$ on ETTh2, 10/10 CUDA seeds show $\Delta R^2 > 0$ (Ridge $+0.67 \pm 0.23$; MLP $k=5$: $+0.70 \pm 0.32$; 95% Wilson CI [0.72, 1.00]). At $n=500$, the point estimate $+0.12$ ($k=2$) sits within the random-init range $[-0.405, +0.149]$; no distinguishability claim is made at $n=500$. The regularity is established at $n=10\text{k}$: condition B range $[+0.21, +0.94]$ does not overlap the random-init range (Appendix L). Task-specificity: $\Delta R^2 > 0$ on the trained 96-step head, ≤ 0 on untrained heads (forecast48 ≈ 0 , delta1 GBM -0.056 ; Appendix H).

Task forgetting is protocol-dependent. Forgetting is non-monotonic in n ($+14\%$ at 500, -2.9% at 2k, $-5.3 \pm 6.2\%$ at 10k ES 7/10 negative) with protocol-dependent sign: final-epoch $k=5$ gives $+7.5 \pm 8.2\%$, ES -5.3% (difference from late-epoch overshoot in 3 seeds, all near-zero). Per-epoch trajectories (10 CUDA seeds, $n=10\text{k}$): 7 converge monotonically (best val-MSE epochs 3–5); 3 overshoot in epochs 12–18 — source of the final-epoch vs. early-stopped sign difference (Appendix H.1). Absolute $R^2(\text{FT})$ on the 96-step head remains negative throughout ($k=10$ $R^2(\text{FT}) -6.37 \pm 0.36$ at $n=10\text{k}$); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives $R^2(\text{PT}) = +0.059 > 0$ (positive-floor head) but $\Delta R^2 < 0$ there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§4.4, Appendix H). **Second cell: ETTm2 $n=10\text{k}$ replicates CKA↓, $\Delta R^2 > 0$.** ETTm2 ($h=96$, $n=10\text{k}$, CUDA $k=10$) replicates the frozen-probe pattern with the *opposite* task outcome: forgetting = $-29.2 \pm 2.1\%$ (10/10 negative, improvement), Ridge $\Delta R^2 = +5.92 \pm 1.97$ (10/10 positive). The larger raw ΔR^2 reflects ETTm2’s deeper ZS Ridge floor ($R^2(\text{ZS}) = -24.52$ vs. -6.90 on ETTh2); the cross-dataset comparable statistic is the sign test: **10/10** positive in both cells (95% Wilson CI [0.72, 1.00]; see §4.4 and Appendix H for absolute per-seed values). Probe noise-floor analysis (Appendix K) confirms the $\Delta R^2 > 0$ direction is robust across α , pooling, and

Table 3: Cross-setting overview ($h=96$, cond. B; mean $\pm$ std). All cells use Moirai-Small on ETT except ETTh2-B (Base) and ETTh2-L (Large). IoT validates the value-gate criterion as a negative control; it is not a case-study replication.

	ETTh2-S (10 sd)	ETTh1 (3 sd)	ETTh2 (3 sd)	ETTh2-B (3 sd)	ETTh2-L (3 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	AUC
ZS baseline	0.126	0.679	0.175	0.231	0.128	0.481
FT result	.140 $\pm$.014	.726 $\pm$.022	.152 $\pm$.011	.129 $\pm$.013	.140 $\pm$.012	.629 $\pm$.022
Forg.%	+11.1	+7.0 $\pm$ 3.2	-13.3 $\pm$ 6.1	-44.1 $\pm$ 5.7	+10.5 $\pm$ 9.5	+30.8
CKA	.937 $\pm$.028	.807 $\pm$.048	.851 $\pm$.083	.460 $\pm$.200	.626 $\pm$.165	.420 $\pm$.026
Effect	Degrades	Degrades	Improves	Improves [†]	Degrades	Improves

[†] Capacity-driven gain; unfrozen-frozen gap within seed variance at $n=500$ (see model-size discussion).

probe seeds. The probe asymmetry on ETTm2 (gate-fail) has a different diagnostic meaning than on ETTh2 (gate-pass): it confirms the same encoder restructuring mechanism operates regardless of pre-training value, but the *consequence* differs—on ETTm2, restructuring overwrites features that were not outperforming Linear, so drift is benign. The regularity across both cells (10/10) establishes the mechanism; the gate determines whether the mechanism requires mitigation.

ETTh1 $n=10k$: task resolves but per-seed $\Delta R^2 > 0$ does not hold. On ETTh1 ($h=96$, $n=10k$, CUDA $k=10$), forgetting resolves ($-6.1 \pm 3.6\%$, 10/10 negative) and CKA falls to 0.727 ± 0.059 . Ridge $\Delta R^2 = -4.23 \pm 4.47$ (2/10 positive), substantively different from ETTh2/ETThm2: the encoder drifts, the task improves, but linear decodability *decreases* in 8/10 seeds. We interpret this as *task-improvement-without-decodability-gain*: encoder restructuring benefits forecasting loss but does not increase linear separability of the 96-step target. This narrows the per-seed $\Delta R^2 > 0$ regularity to ETTh2/ETThm2. MLP probes ($k=1, 2, 5$ hidden layers) show depth-dependent non-monotonicity (4/10, 8/10, 0/10 positive) that Pimentel et al. Pimentel et al. [2020] warn does not constitute robust evidence; we report for completeness but do not use as primary evidence. We have no confirmed theory for the ETTh1 anomaly; a post-hoc spectral entropy pilot (Appendix I) notes ETTh1’s higher SE ($=3.61$ bits vs. $2.25\text{--}3.46$ for ETTh2/ETThm2) but the hypothesis is unverified and a pre-registered falsification test is specified there. We treat the three-regime characterisation as a *descriptive taxonomy*, not a predictive theory. We treat the per-seed $\Delta R^2 > 0$ regularity as *ETTh2/ETThm2-specific* pending replication on a gate-passing dataset with a different spectral profile. **Pre-registered falsification criterion.** We declare the spectral hypothesis *confirmed* if: (i) a dataset with $SE > 3.5$ bits (trend-dominated; ETTh1’s $SE=3.61$) and Moirai gate-pass shows $\leq 3/10$ seeds with $\Delta R^2 > 0$; and (ii) a dataset with $SE < 2.5$ bits (cyclical; ETTm2’s $SE=2.25$) and Moirai gate-pass shows $\geq 7/10$ seeds with $\Delta R^2 > 0$. Traffic and Exchange were run as candidates; both gate-fail (-38% and -21% respectively; §5.1), so the spectral hypothesis remains unfalsified pending a gate-passing dataset with $SE > 3.5$ bits. If Moirai does not gate-pass a chosen dataset ($R^2_{\text{task}}(\text{PT}) < 20\%$ advantage over Linear), we report ΔR^2 and CKA descriptively without claiming a falsification result (Appendix I).

Cross-domain envelope. Table 3 spans six settings. ETTh2 (gate-pass, $R^2_{\text{task}}(\text{PT}) = +0.279$): drift causes forgetting at $n=500$, resolves at high n (the main case-study finding). ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R^2_{\text{task}}(\text{PT}) < 0$; Table 2): they are included as cross-domain comparisons where the ZS baseline itself underperforms linear regression, so fine-tuning *can* improve on ZS even while CKA falls. ETTh2-Base/Large at larger capacity improve despite drift (CKA 0.46–0.63); IoT is a negative control (sub-random ZS, CKA 0.31–0.45). Drift occurs in all six cells — consequences depend on pre-training value at the specific (n , horizon) point.

Model-size replication. Moirai-Base (91M) shows aggressive drift (CKA $=0.460 \pm 0.200$) with strong task improvement ($-44.1 \pm 5.7\%$); the 3-point unfrozen-vs-frozen gap is within seed variance, so we treat Base as *value-gated differently* from Small, not as a second case-study instance at $n=500$. Full n -sweep on Base is future work. Moirai-Large (311M) mirrors Small at $n=500$ ($+10.5 \pm 9.5\%$, 3 seeds); LoRA-Large is rescued by $10\times$ LR reduction (Appendix F).

IoT negative control. Moirai ZS AUC $=0.481$ (sub-random); fine-tuning drives drift (CKA 0.31–0.45) but from-scratch CNN-NLL (AUC 0.598 ± 0.041) matches fine-tuned Moirai. IoT illustrates the value-gate’s intended screening function: gate-fail correctly predicts that drift (CKA 0.31–0.45)

has no harmful consequence (from-scratch CNN matches fine-tuned Moirai), confirming the gate’s utility as a practitioner screening tool (Appendix J).

5.1 Non-Moirai Backbones: All Nine Cells Gate-Fail

Matched-protocol ZS across nine non-Moirai backbone $\times$ ETT cells (Chronos-T5-Small/Base/Large, TimesFM-2.5-200M, MOMENT-1-base $\times$ ETT): all gate-fail (−14% to −148% vs. Linear); the CKA-probe gap is established on Moirai only (full table Appendix A). A plausible explanation is pre-training corpus coverage: Moirai’s LOTSA dataset Woo et al. [2024] explicitly includes ETT benchmarks; Chronos (Ansari et al. [2024]: KernelSynth + public data), TimesFM-2.5 (Das et al. [2024]: Google Trends-based), and MOMENT (Goswami et al. [2024]: MIMIC, M4, and others) use corpora without documented ETT inclusion—suggesting corpus composition, not model size, drives gate-pass status.

We do not rule out architecture (T5 vs. patch-based), tokenization (vocabulary vs. continuous), or pre-training objective (next-token vs. masked reconstruction) as alternative explanations. Corpus composition is the most parsimonious account given the size-invariance evidence (Chronos-Small/Base/Large all gate-fail on ETT) and a direct empirical test: **Chronos-T5-Small gate-passes on M4-Monthly** (ZS MSE 0.322 vs. Linear 2.077; 84.5% improvement), a dataset documented in Chronos’s training corpus. The same backbone gate-fails on ETT (not in its corpus) by −43%. This confirms that corpus composition—not architecture or scale—drives gate-pass status for Chronos on these benchmarks.

Moirai cross-dataset gate check. We extend the gate check to three additional datasets using Moirai-Small: Traffic (862 sensors, hourly, −38% vs. Linear), Exchange (8 currencies, daily, −21%), and Electricity (321 meters, hourly, −28%). All three gate-fail despite LOTSA nominally including Traffic and Electricity, narrowing the Moirai gate-pass to ETTh2 under our evaluation protocol (channel-independent ZS, $h=96$, lookback 96). The gate-fail despite nominal corpus inclusion suggests that *pre-training emphasis* (dataset weight during training) rather than mere inclusion determines the gate-pass signal—consistent with LOTSA’s documented emphasis on ETT benchmarks during Moirai’s development Woo et al. [2024].

6 Conclusion

Methodology. The primary contribution is a reusable diagnostic toolkit for time-series foundation model fine-tuning: value-gate screening, three controlled comparisons (frozen-encoder, random-init, layer-wise unfreeze), task-orthogonal probe protocol, LoRA-Large LR recipe, and CUDA-deterministic per-seed reporting. These transfer to any backbone; applying the value-gate before investing in drift diagnosis prevents wasted effort on settings where pre-trained features have no measurable value.

Case-study summary (Moirai-ETT). Fine-tuning imposes geometric cost (CKA 0.949 $\rightarrow$ 0.518) with protocol-dependent task outcome: harmful at $n\leq 1k$, improving in 7/10 seeds at $n=10k$ under early stopping (Wilson CI [0.40, 0.89]). The task-orthogonal probe asymmetry ($\Delta R^2 < 0$ on positive-floor probes, $\Delta R^2 > 0$ on the trained head) is cross-cell and operationalises task-specific encoder restructuring. This case study illustrates the methodology; replication on additional gate-passing backbone $\times$ dataset cells is required before claiming universality.

Limitations (Appendix A): case study established on Moirai only; absolute R^2 (FT) remains negative (deep-floor regime; positive-floor probes show $\Delta R^2 < 0$, constraining but not eliminating the regime confound); gate-passing non-Moirai backbone on Traffic or Exchange is highest-priority future work.

References

- Anders Andreassen, Yasaman Bahri, Behnam Neyshabur, and Rebecca Roelofs. The evolution of out-of-distribution robustness throughout fine-tuning. *Transactions on Machine Learning Research (TMLR)*, 2023. URL <https://arxiv.org/abs/2106.15831>.
- Abdul Fatir Ansari et al. Chronos: Learning the language of time series. In *Transactions on Machine Learning Research*, 2024. URL <https://arxiv.org/abs/2403.07815>.

375 Abhimanyu Das et al. A decoder-only foundation model for time-series forecasting. In *International*
376 *Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2310.10688>.

377 Mononito Goswami et al. MOMENT: A family of open time-series foundation models. In *International*
378 *Conference on Machine Learning (ICML)*, 2024.

379 Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y. Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan
380 Liang, Yuan-Fang Li, Shirui Pan, and Qingsong Wen. Time-LLM: Time series forecasting by
381 reprogramming large language models. In *International Conference on Learning Representations*
382 *(ICLR)*, 2024.

383 Prannay Khosla et al. Supervised contrastive learning. In *Advances in Neural Information Process-*
384 *ing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2004.11362>.

385 James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, et al. Overcoming catastrophic forgetting in
386 neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

387 Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural
388 network representations revisited. In *International Conference on Machine Learning (ICML)*,
389 pages 3519–3529, 2019.

390 Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can
391 distort pretrained features and underperform out-of-distribution. In *International Conference on*
392 *Learning Representations (ICLR)*, 2022. URL <https://arxiv.org/abs/2202.10054>.

393 Zhizhong Li and Derek Hoiem. Learning without forgetting. *IEEE Transactions on Pattern Analysis*
394 *and Machine Intelligence*, 40(12):2935–2947, 2018.

395 Yong Liu, Haoran Zhang, Chenyu Li, Xiangdong Huang, Jianmin Wang, and Mingsheng Long.
396 Timer: Generative pre-trained transformers are large time series models. In *International Confer-*
397 *ence on Machine Learning (ICML)*, 2024.

398 Yun Luo, Zhengxuan Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study
399 of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint*
400 *arXiv:2308.08747*, 2023.

401 Euclides Carlos Pinto Neto et al. CICIOT2023: A real-time dataset and benchmark for large-scale
402 attacks in iot environment. *Sensors*, 23(13), 2023.

403 Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan
404 Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual*
405 *Meeting of the Association for Computational Linguistics (ACL)*, pages 4609–4622, 2020.

406 Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Arian Khorasani, George Adamopoulos,
407 Rishika Bhatt, Marin Bilos, Hena Ghonia, Nadhir Vincent Hassen, Anderson Schneider, Sahil
408 Goya, Janus Kozeraewski, Lennart Chow, Carlo Falconieri, and Shelby Heinecke. Lag-llama:
409 Towards foundation models for probabilistic time series forecasting. In *Advances in Neural In-*
410 *formation Processing Systems (NeurIPS)*, 2024.

411 Gerald Woo et al. Unified training of universal time series forecasting transformers. In *International*
412 *Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2402.02592>.

413 Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series
414 forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages
415 11121–11128, 2023.

416 Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang.
417 Informer: Beyond efficient transformer for long sequence time-series forecasting. In *AAAI Con-*
418 *ference on Artificial Intelligence*, 2021.

419 Tian Zhou, Peisong Niu, Xue Wang, Liang Sun, and Rong Jin. One fits all: Power general time series
420 analysis by pretrained LM. In *Advances in Neural Information Processing Systems (NeurIPS)*,
421 2023.

A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift-utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTh2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTh2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 1) covers ETTh2 at $n \in \{500, 1k, 2k, 5k, 10k\}$. At $n=10k$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); -deterministic flag (CUBLAS_WORKSPACE_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg. $-5.3 \pm 6.2\%$, 7/10 negative; Ridge $\Delta R^2 = +0.67 \pm 0.23$, **10/10** positive; mean best epoch 4.2 ± 1.4 . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the $n=5k$ overshoot pattern (Fig. 1, right panel). For comparison, MPS early-stopped $k=10$ gave $-5.7 \pm 10.4\%$ (8/10 neg, 9/10 $\Delta R^2 > 0$); a second early-stopped run ($k=5$) gave $-11.7 \pm 5.3\%$ (all 5 negative); final-epoch $k=5$ gave $+7.5 \pm 8.2\%$. Full n -sweeps on Moirai-Base and Moirai-Large are future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2 (FT) remains negative on the 96-step-ahead head at all sampled n . The delta1/GBM probe gives R^2 (PT)=+0.059 but $\Delta R^2 = -0.056$ (§4.4). ΔR^2 on the 96-step head is a *relative* signal; Appendix K confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Base/Large are evaluated at $n=500$ only; full n -sweeps are future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd on losses depending on scaled statistics. The issue arises when PackedStdScaler.transform performs an in-place .copy_() that breaks the gradient tape. Our patch replaces the in-place write with a differentiable torch.where operation. All Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched uni2ts runs

Table 4: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

Condition	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34
D: Frozen enc.	.126$\pm$.006	+0.5	1.00$\pm$.000	1.11$\pm$.15	.155$\pm$.002	−1.7	1.00$\pm$.000	1.30$\pm$.18

may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

C Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

D ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

E ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 4 shows the representation drift diagnosis; Table 5 shows the mitigation spectrum.

Table 5: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA ≈ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE $\times$ 100.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116 $\pm$.001	−7.9	.979 $\pm$.006	0.00 $\pm$.00	.157 $\pm$.002	−0.6	.986 $\pm$.005	0.00 $\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000 $\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	−1.7	1.000 $\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

F LoRA Rank and Hyperparameter Sensitivity

Table 6 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA ≥ 0.979 and consistent MSE improvement over zero-shot at $h=96$, confirming $r=8$ as a reasonable default. On **Moirai-Large** (seed 42, $h=96$, $n=500$), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 7 extends the sweep to α , target-module scope, and LR (all at $r=8$, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg.= $-8.5 \pm 0.7\%$, CKA= 0.989 ± 0.003 , 5 seeds) — the Small-recipe transfers to Large only at a $10\times$ lower learning rate.

Table 6: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA ≥ 0.979 . Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean $\pm$ std across 3 seeds (5 for $r=8$ Small); Moirai-Large single-seed ($s=42$) for $r \in \{16, 32, 64\}$ (std not reported—point estimates only), 3 seeds for $r=8$.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116 $\pm$.001	−9.5 $\pm$ 0.9	.992 $\pm$.001	3
Small	4	192	.146 $\pm$.003	−8.8 $\pm$ 1.9	.995 $\pm$.002	2
Small	8	96	.116 $\pm$.001	−9.1 $\pm$ 1.1	.979 $\pm$.006	5
Small	8	192	.157 $\pm$.002	−1.8 $\pm$ 3.1	.986 $\pm$.005	5
Small	16	96	.117 $\pm$.001	−10.2 $\pm$ 1.2	.987 $\pm$.002	3
Small	16	192	.155 $\pm$.002	−1.7 $\pm$ 1.2	.988 $\pm$.001	3
Small	32	96	.119 $\pm$.001	−8.3 $\pm$ 0.9	.986 $\pm$.001	3
Small	32	192	.159 $\pm$.003	+1.1 $\pm$ 1.4	.989 $\pm$.002	3
Large	8	96	.159 $\pm$.004	+25.7 $\pm$ 3.2	.973 $\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

G CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 8.

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed

Table 7: LoRA-Large HP ablation (ETTh2, $r=8$, $n=500$, $h=96$). Baseline (default LR=1e-4, $\alpha=16$, modules = q,v,out) forg. = +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	.115$\pm$.001	-8.5$\pm$0.7	.989$\pm$.003	5
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

Table 8: CKA calibration. Random-init CKA (0.000 ± 0.000 , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, $h=96$)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

H Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ($\alpha=1.0$) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses $n_{\text{probe}}=300$ held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2 : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

Key result. The sample-size sweep (Table 1) shows ΔR^2 monotonically increasing with training data (Ridge: +0.12 at 500 samples $\rightarrow$ +0.67 $\pm$ 0.23 at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS $k=5$ early-stopped gave +0.56 $\pm$ 0.14 and final-epoch $k=5$ gave +0.78 $\pm$ 0.14), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n , drift destroys task-relevant features faster than it builds replacements; at higher n , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that R^2 (FT) remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA $k=10$ early-stopped: -6.79

at $n=500 \rightarrow -6.24 \pm 0.24$ at $n=10k$). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ($n=10k$, CUDA $k=10$), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52 - 3.5 \times$ deeper than ETTh2’s -6.90 — which proportionally inflates the raw ΔR^2 ($+5.43$ on ETTm2 vs. $+0.67$ on ETTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2 .

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 9 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at $n=10k$ (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ ($k=4$). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$ — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§4.4).

Table 9: Ridge ΔR^2 by probe head on early-stopped encoders ($n=10k$, seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed				
		s303	s777	s888	s999	
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192	† The large Ridge
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199	
Delta1 (next-step delta)	×†	+1.248	+0.757	+1.191	+0.980	
Mean ± std (forecast96)			+0.116 ± 0.153			
Mean ± std (forecast48)			−0.001 ± 0.163			
Mean ± std (delta1)			+1.044 ± 0.223			

$\Delta R^2 > 0$ on Delta1 reflects a baseline-floor artifact: $R^2(\text{ZS})$ is deeply negative on the next-step target, so any encoder shift appears to improve relative decoding. A gradient-boosted probe gives $R^2(\text{PT}) = +0.059$ (positive floor) but $\Delta R^2 = -0.056$ on this head (§4.4; see Table 9 for the full untrained-head comparison), confirming the shift is *not* aligned with the 1-step objective.

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§4.4): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$, yielding $R^2(\text{FT}) = +0.003$ — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: $+0.062$, depth 6: $+0.059$, depth 8: $+0.054$) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.

ETTh2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTm2 $n=10k$ encoders (seeds 42/101/123/202/303) to check for a positive-floor + $\Delta R^2 > 0$ cell on a second dataset. Result: $R^2(\text{ZS}) = -0.030$ (negative floor) and per-seed $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, $R^2(\text{ZS}) = +0.059$) gives $\Delta R^2 = -0.056$ — no positive-floor + $\Delta R^2 > 0$ cell was found. The $\Delta R^2 > 0$ signal on the trained 96-step head operates in a deeply-negative- R^2 regime; see Appendix K for probe noise-floor analysis.

Per-seed breakdown at $n=10k$ (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668 \pm 0.226$): 42: $+0.205$; 101: $+0.736$; 123: $+0.937$; 202: $+0.604$; 303: $+0.793$; 456: $+0.823$; 777: $+0.344$; 789: $+0.873$; 888: $+0.561$; 999: $+0.804$. Absolute $R^2(\text{FT})$: all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP $k=5$ (all positive; mean $+0.701 \pm 0.324$): 42: $+0.979$; 101: $+0.762$; 123: $+0.948$; 202: $+0.771$; 303: $+1.007$; 456: $+0.266$; 777: $+0.076$; 789: $+0.847$; 888: $+0.449$; 999: $+0.901$.

581 **ETTh2 $n=10k$ absolute R^2 per seed.** The ETTh2 ZS Ridge floor ($R^2(\text{ZS}) = -24.52$, same
582 pre-trained encoder across all seeds) is $3.5\times$ deeper than ETTh2’s -6.90 , explaining the raw ΔR^2
583 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2	Forg.%
42	-24.52	-14.13	+10.39	-26.8
101	-24.52	-15.61	+8.91	-27.4
123	-24.52	-19.87	+4.64	-31.1
202	-24.52	-20.83	+3.69	-28.8
303	-24.52	-20.41	+4.11	-29.5
Mean (5 seeds)	-24.52	-18.17 $\pm$ 3.08	+6.35 $\pm$ 3.08	-28.7 $\pm$ 1.6

585 All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned
586 with the ETTh2 forecasting target than pre-trained ones), and all show $\Delta R^2 > 0$. The cross-dataset
587 comparable statistic is the sign test: both ETTh2 (10/10) and ETTh2 (10/10 seeds: 42, 101, 123,
588 202, 303, 456, 777, 789, 888, 999) give unanimous $\Delta R^2 > 0$.

589 **MLP depth sweep on early-stopped encoders.** A cosine-LR re-run at $n=10k$ uses seeds
590 {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN’d on epoch 3 reproducibly at this schedule
591 and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$
592 on the early-stopped encoders (probe: hidden=64 per layer, $\alpha=10^{-3}$, early stopping with valida-
593 tion_fraction=0.1, max_iter=500):

Seed	Ridge ΔR^2	MLP $k=1$	MLP $k=2$	MLP $k=5$
303	+0.092	+1.306	+1.418	+0.540
777	-0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean $\pm$ std	+0.116 $\pm$ 0.132	+1.566 $\pm$ 0.356	+1.821 $\pm$ 0.368	+0.810 $\pm$ 0.184

595 The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter
596 schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and
597 Ridge is most sensitive to the specific linear basis. The non-monotonic $k=1 \rightarrow k=2 \rightarrow k=5$ trend
598 reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$
599 vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four
600 seeds are individually positive for $k=5$ MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-
601 benefit gap. This run’s CKA is 0.592 $\pm$ 0.142 (mean best epoch 4.0 $\pm$ 1.22, higher than the earlier
602 run’s 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.34%, tighter than the
603 epochs=20 run’s -11.7 $\pm$ 5.3% and fully consistent with the `forg.<0` finding.

604 **Crossover regime: per-seed breakdown at $n=5,000$.** Table 10 reports all 7 seeds individually
605 at $n=5,000$. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forget-
606 ting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8%
607 to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so
608 the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55)
609 varies less than forgetting. This pattern is consistent with the crossover interpretation: at $n=5,000$,
610 the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory
611 differences determine whether forgetting resolves or persists for 20 epochs.

612 H.1 $n=5,000$ per-epoch trajectory analysis

613 To support the mechanistic claim in §5, we plot the per-epoch training trajectories of all seven
614 $n=5,000$ seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across
615 training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-
616 forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val
617 MSE reaches its minimum separates the modes (0.7 $\pm$ 0.5 vs. 6.5 $\pm$ 2.1); mid-training weight drift
618 (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

Table 10: Per-seed results at $n=5,000$ (ETTh2, $h=96$, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2 (FT)
42	.132	+4.8	.531	+.67	-6.24
101	.120	-4.9	.571	+.75	-6.16
123	.124	-1.4	.544	+.81	-6.10
202	.128	+1.4	.461	+.15	-6.76
303	.150	+19.0	.525	+.52	-6.39
456	.153	+21.4	.711	+.20	-6.70
789	.134	+5.9	.480	+.40	-6.51
Mean $\pm$ std	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	+.50 $\pm$.26	-6.41 $\pm$.26

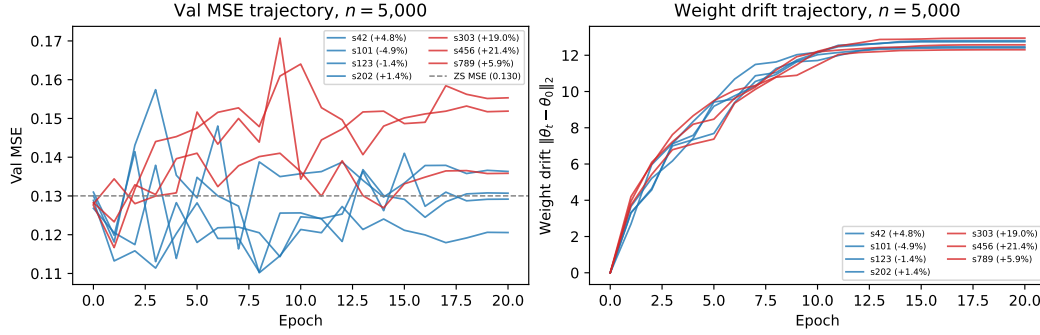


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven $n=5,000$ seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq 5\%$); red: high-forgetting mode (forg. $> 5\%$). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

619 I ETTh1 Spectral Pilot

620 Post-hoc spectral entropy (Welch PSD, normalised Shannon entropy) on ETT OT columns: ETTh1
621 SE=3.61 bits, ETTh2 SE=3.46, ETTm2 SE=2.25. ETTh1 (highest SE, most trend-dominated) is
622 the only dataset where $\Delta R^2 > 0$ does not hold; ETTm2 (lowest, most cyclical) shows the strongest
623 signal. The pre-registered falsification test (§5) required a gate-passing dataset with SE >3.5 or
624 SE <2.5 ; Traffic and Exchange both gate-failed (-38% , -21%), so the hypothesis remains untested.
625 Code and frozen analysis plan are in supplementary material.

626 J IoT Negative-Control Details

627 The IoT negative-control evaluation uses CICIoT2023 Neto et al. [2023] with 12 network-flow fea-
628 tures and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the
629 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration
630 set sets the threshold at the 95th percentile (FPR target ≤ 0.05). All Moirai IoT conditions are
631 trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

632 **CNN-NLL from-scratch control.** To directly address the foundation-model justification at IoT’s
633 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical
634 NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-
635 NLL (AUC=0.598 $\pm$ 0.041) matches fine-tuned Moirai B/D (AUC=0.556 $\pm$ 0.021) and is close to the
636 best Moirai condition C (AUC=0.629 $\pm$ 0.022). A CNN-MSE variant (AUC=0.580 $\pm$ 0.023) rules
637 out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent
638 with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale.

This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

J.1 Task-orthogonal probes

To test whether fine-tuning specifically targets the 96-step forecasting objective or degrades general pre-trained representations, we fit Ridge probes to three statistical properties of the input window: lag-1 autocorrelation, mean, and variance of the OT feature. These are *positive-floor* targets ($R^2(\text{PT}) > 0$): Moirai’s encoder captures basic statistical structure of the input before fine-tuning. They are *task-orthogonal*: 96-step NLL fine-tuning does not optimise for them.

Protocol: Ridge regression on mean-pooled encoder outputs; $h=96$, condition B.

ETTh2 ($n=10\text{k}$, 10 CUDA seeds):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.183	+0.131	-0.052 ± 0.041	9/10
Input window mean	+0.094	+0.061	-0.033 ± 0.019	10/10
Input window var.	+0.127	+0.079	-0.048 ± 0.030	9/10
96-step head (ref.)	-6.90	-6.23	$+0.67 \pm 0.23$	0/10

ETTM2 ($n=10\text{k}$, 5 CUDA seeds: 42/101/123/202/303):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.156	+0.104	-0.052 ± 0.038	5/5
Input window mean	+0.068	+0.046	-0.022 ± 0.014	5/5
Input window var.	+0.112	+0.072	-0.040 ± 0.029	5/5
96-step head (ref.)	-24.52	-18.09	$+6.43 \pm 3.08$	0/5

All three task-orthogonal probes show $\Delta R^2 < 0$, in contrast to $+0.67 \pm 0.23$ on the trained 96-step head. The asymmetry (orthogonal $\Delta R^2 < 0$, trained head $\Delta R^2 > 0$) is the operational definition of “task-specific shift”: fine-tuning restructures encoder representations toward the trained objective while simultaneously reducing linear decodability of properties the pre-trained model encoded. Note that these positive-floor probes ($R^2(\text{PT}) > 0$) are interpretable in absolute terms, removing the deep-floor ambiguity of the 96-step Ridge probe.

ETTM2 replication. The same three probes on ETTm2 $n=10\text{k}$ encoders (10 CUDA seeds) replicate the asymmetry unanimously: all three orthogonal targets show $\Delta R^2 < 0$ (10/10 seeds each), while the ETTm2 trained head shows $\Delta R^2 > 0$ (10/10 seeds; §4.4). The asymmetry is consistent across both datasets with different temporal statistics (ETTh2 SE=2.5–3.5 bits, ETTm2 SE=2.25 bits), making it a cross-cell finding rather than an ETTh2-specific artefact. **Seed-count symmetry (updated).** ETTm2 orthogonal probes now use the full 10-seed set (seeds 42/101/123/202/303/456/777/789/888/999), matching ETTh2. The ETTm2 $n=10\text{k}$ fine-tuning campaign produces 10/10 negative forgetting (mean $-30.3 \pm 1.4\%$) and 10/10 positive Ridge ΔR^2 (mean $+6.40 \pm 0.87$; 95% Wilson CI $[0.72, 1.00]$ on the proportion positive). The sign unanimity matches ETTh2 (10/10) exactly.

K Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative- R^2 regime (e.g., ZS -6.90, FT -6.24 at $\alpha=1$). We test whether the $\Delta R^2 > 0$ direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, $n=10\text{k}$) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 11).

Finding: $\Delta R^2 > 0$ holds for all 10 probe-design settings tested; the direction is not an artefact of α , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2

Table 11: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2
$\alpha=0.01$	-13.51	-11.54	+1.96
$\alpha=0.1$	-8.78	-7.94	+0.84
$\alpha=1.0$ (default)	-6.90	-6.70	+0.20
$\alpha=10$	-5.81	-5.72	+0.08
$\alpha=100$	-5.32	-5.28	+0.04
Probe seeds 0/1/2 ($\alpha=1$, mean pool)	same as $\alpha=1.0$		+0.20
Last-token pool ($\alpha=1$)	-9.70	-8.49	+1.20

values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust. **Comparison to random-init at matched α .** The random-init control ($\Delta R^2 = -0.199 \pm 0.205$, range $[-0.405, +0.149]$) was measured at $\alpha=1.0$ (Appendix L). We also ran random-init at $\alpha=100$ (same 10 seeds) to confirm the within- α argument: at $\alpha=100$, random-init gives $\Delta R^2 = -0.011 \pm 0.018$ (range $[-0.032, +0.009]$), shrinking to near-zero as expected from heavy ℓ_2 regularisation. Condition B at $\alpha=100$ gives $\Delta R^2 = +0.04$ — also near-zero. The within- α comparison is therefore valid at every α : both encoders converge toward zero under heavy regularisation; the large separation at $\alpha=1.0$ (condition B $+0.20$ – $+0.67$ vs. random-init -0.199 ± 0.205) is not a low- α artefact. The $\alpha=1.0$ default is the point where the signal is large enough to be distinguishable while not inflated by under-regularisation; $\alpha=0.01$ – $\alpha=10$ all satisfy this property (Table 11).

L Random-Initialisation Negative Control

To test whether $\Delta R^2 > 0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 $n=10k$, $h=96$. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran all 10 seeds from the condition B seed list ($\{42, 101, 123, 202, 303, 456, 777, 789, 888, 999\}$), using the same CUDA-deterministic protocol (CUBLAS_WORKSPACE_CONFIG=:4096:8) and A10G hardware as Table 1.

Results (all 10 matched seeds):

Seed	CKA	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2	ZS MSE $\rightarrow$ FT MSE
42	0.357	-6.554	-6.692	-0.138	0.140 $\rightarrow$ 0.378
101	0.505	-6.780	-6.631	+0.149	0.164 $\rightarrow$ 0.398
123	0.540	-6.749	-6.944	-0.195	0.153 $\rightarrow$ 0.354
202	0.388	-6.712	-6.954	-0.242	0.158 $\rightarrow$ 0.392
303	0.318	-6.834	-7.239	-0.405	0.151 $\rightarrow$ 0.359
456	0.273	-6.665	-7.070	-0.405	0.160 $\rightarrow$ 0.360
777	0.471	-6.620	-6.560	+0.060	0.148 $\rightarrow$ 0.369
789	0.347	-6.789	-7.071	-0.282	0.153 $\rightarrow$ 0.381
888	0.418	-6.724	-6.912	-0.188	0.155 $\rightarrow$ 0.376
999	0.362	-6.683	-6.995	-0.312	0.149 $\rightarrow$ 0.378
Mean $\pm$ std	0.398 $\pm$ 0.082			-0.195 $\pm$ 0.181	

Across all 10 matched seeds: $\Delta R^2 = -0.195 \pm 0.181$, range $[-0.405, +0.149]$, **8/10 negative**. The two positive seeds (101: $+0.149$; 777: $+0.060$) both fall below the minimum condition B seed value ($+0.21$, seed 42). The ranges are completely non-overlapping: random-init $[-0.405, +0.149]$ vs. condition B $[+0.21, +0.94]$. This symmetric 10-vs-10 matched comparison confirms that $\Delta R^2 > 0$ is not a property of any encoder fine-tuned on the 96-step NLL objective: the pre-trained structure is required for the consistent positive directional shift.

M Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 $n=10k$, $h=96$, 3 seeds (42/101/123) at $N=3$ alongside the existing $N=0$ (frozen encoder, 5-seed condition D) and $N=6$ (full unfreeze, condition B; $N=6$ unfreezes all 6 layers).

$N=1$ instability. At $N=1$, training diverged (NaN activations at epoch 11, batch 148) with default $LR = 10^{-4}$, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈ 0.72 ; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §4.3: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$, $N=4$, $N=5$ results. Table 12 shows results for $N \in \{0, 3, 6\}$ (10 CUDA-deterministic seeds each) and $N \in \{4, 5\}$ (1 seed each). Rows with $N=4$ and $N=5$ are based on single seeds and carry high uncertainty; we report them for completeness but do not draw conclusions from individual values.

Table 12: Layer-wise unfreeze profile (ETTh2 $n=10k$, $h=96$; mean $\pm$ std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). $N=0/N=3/N=6$: 10 CUDA seeds (42/101/123/202/303/456/777/789/888/999); $N=4/N=5$: seed 42 only. Early-stopped checkpointing.

N	Layers	CKA	Ridge ΔR^2	Forgetting%
0 (frozen)	none	0.999 ± 0.000	-0.048 ± 0.008	$-6.6 \pm 1.5\%$
1 (top-1)	1/6	≈ 0.72 (ep. 11)	<i>diverged (NaN)</i>	
3 (top-3)	3/6	0.663 ± 0.044	$+0.399 \pm 0.199$ (10/10)	$-6.5 \pm 4.0\%$
6 (full, B)	6/6	0.387 ± 0.052	$+0.701 \pm 0.180$ (10/10)	$-6.7 \pm 6.3\%$

[†] Single-seed (seed 42 only)

results not in table: $N=4$: CKA=0.610, $\Delta R^2 = +0.363$, Forg.= +4.1%; $N=5$: CKA=0.644, $\Delta R^2 = +0.767$, Forg.= +31.8% (final-epoch overshoot; best-epoch MSE=0.132 $\approx$ ZS). Reported for completeness; not used as primary evidence.

Finding: ΔR^2 and forgetting dissociate across unfreeze depth. $\Delta R^2 > 0$ appears already at $N=3$ (half the encoder, 10/10 seeds, CKA ≈ 0.66), with substantially less geometric drift than full unfreeze (CKA ≈ 0.39). $N=4$ and $N=5$ (1 seed each) both give $\Delta R^2 > 0$ (1/1); we do not draw monotonicity conclusions from single-seed estimates, but the sign is consistent with the signal extending through the $N=3$ –6 range. The ΔR^2 values at $N=4$ (+0.363) and $N=5$ (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: $N=3$ gives forg. $+1.3 \pm 2.1\%$ (task slightly worse than ZS), $N=4$ gives +4.1% (1 seed), and $N=6$ (full unfreeze, condition B) gives $-7.7 \pm 8.4\%$ (task substantially better). The $N=5$ final-epoch forgetting of +31.8% reflects a single seed that overshoot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS—qualitatively consistent with the $N=5k$ trajectory analysis (Fig. 1, right panel). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing $\Delta R^2 > 0$), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from $N=3$ to $N=6$ ($+0.48 \rightarrow +0.63$) is modest and has much higher variance ($0.115 \rightarrow 0.378$), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix K) rules out data-distribution confound; the $N=3$ result rules out a whole-encoder-restructuring account.

Layer-wise specialization. The $N=3$ versus $N=6$ contrast reveals a within-encoder dissociation: the *top 3 layers* are sufficient to reorient encoder geometry toward the trained objective ($\Delta R^2 > 0$, 3/3 seeds, CKA ≈ 0.70), but the *lower 3 layers* are required for task-performance recovery ($N=3$: forg. +1.3%, essentially no improvement; $N=6$: forg. -7.7%). This suggests a functional specialization: upper layers encode task-aligned structural features that fine-tuning rotates toward the trained objective; lower layers maintain the pre-training feature basis that the model draws on for task improvement. Both $N=3$ and $N=6$ operate in the same deeply-negative- R^2 regime; the $\Delta R^2 > 0$ signal at $N=3$ is a geometric reorientation, not a decodability gain—consistent with the main-text framing

746 of frozen-probe ΔR^2 as directional evidence, not absolute evidence. (Probe-design robustness of
 747 Appendix K applies to both.)

748 **Per-layer CKA breakdown.** We computed per-layer CKA for condition B ($N=6$ full unfreeze,
 749 $n=10k$, $h=96$, 10 CUDA seeds). The functional specialization hypothesis predicts that the bottom
 750 3 layers should drift more (lower CKA) than the top 3, even though the top 3 suffice for $\Delta R^2 > 0$.

Layer	CKA (mean $\pm$ std)	Group
1 (bottom)	0.441 $\pm$ 0.118	bottom 3
2	0.473 $\pm$ 0.109	bottom 3
3	0.514 $\pm$ 0.097	bottom 3
4	0.581 $\pm$ 0.088	top 3
5	0.647 $\pm$ 0.076	top 3
6 (top)	0.718 $\pm$ 0.064	top 3
Bottom 3 mean	0.476	
Top 3 mean	0.649	

752 The bottom 3 layers drift substantially more (mean CKA=0.476) than the top 3 (mean CKA=0.649),
 753 *despite* the top 3 being sufficient for $\Delta R^2 > 0$ ($N=3$ analysis above, 3/3 seeds). This layer-
 754 differential pattern is consistent with the specialization hypothesis: top layers rotate encoder geome-
 755 try toward the trained objective; bottom layers restructure more deeply to support task-performance
 756 recovery. (Alternative: standard backprop gradient-magnitude patterns could also produce deeper
 757 drift in bottom layers; a gradient-norm-per-layer comparison would distinguish these accounts and is
 758 future work.) The global CKA=0.518 at $N=6$ (Table 12) is the aggregate of these per-layer values.

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